

Distribution-Matched Pair Classification for Cross-Geo Product Grouping

NI-ADM 2026 – GLAMI Product Matching Challenge

1 Introduction

Cross-geo product matching requires identifying identical products spread across listings from different countries, where titles may be in different languages, images may differ in crop and background, and prices are expressed in different currencies. The GLAMI dataset [2] contains approximately 1.3M fashion items from multiple European markets, labelled with 217k distinct product identities. The two competition tasks are (1) binary detection of duplicate-containing groups (Phase 1) and (2) hard clustering of all items into product groups (Phase 2), evaluated by pairwise F1.

The core difficulty is the gap between a retrieval model that must cover all true matches and a precision-oriented classifier that must not merge distinct products. Two additional challenges distinguish this dataset from standard entity matching: (i) the same product may have a Czech title in one listing and a Romanian title in another, making lexical signals useless, and (ii) the inference candidate distribution is much narrower than the training distribution, causing off-the-shelf rerankers trained on the full pair distribution to be severely miscalibrated.

Our system addresses both challenges: multimodal contrastive fine-tuning provides language-agnostic embeddings, and a distribution-matched pair classifier is trained on exactly the similarity range seen at inference. We then apply the same principle a second time, training a separate classifier for the medium-similarity range with negative sampling that matches the inference positive prior, recovering cross-geo pairs the high-similarity model alone cannot reach.

2 Related Work

Zalando end-to-end product matching. The most directly related work is Tóth et al. [1], who present an industry-scale system for cross-geo duplicate detection in fashion e-commerce. Their approach projects pretrained image and text encoders into a shared space via contrastive learning and outperforms both single-modality systems and large pretrained models such as CLIP. A human review loop is used in production to achieve near-perfect precision. Our work differs in two important respects. First, we target a fully automated system without human review, which forces the classifier to be more precise. Second, and more fundamentally, we identify and correct the *calibration mismatch* that arises in the retrieve-then-rerank pipeline: Tóth et al. do not discuss training the reranker on the same similarity region it will encounter at inference, whereas we show this is the single most important factor in our ablation: an identical XGBoost trained on the full pair distribution fails catastrophically ($F1 < 0.10$), an

absolute gap of more than 80 F1 points relative to the distribution-matched variant (Section 6).

Entity matching with language models. DITTO [3] and DeepMatcher [4] apply pre-trained transformers to textual entity matching. They achieve strong results in English, but neither addresses the cross-lingual or multimodal setting required here.

Multimodal contrastive learning. CLIP [5] demonstrated that joint image-text contrastive training produces powerful shared representations. We adopt the same in-batch InfoNCE objective in a supervised, domain-adapted setting: fine-tuning separate text and image towers on same-label item pairs to align cross-geo product representations in the GLAMI fashion domain.

Approximate nearest-neighbour reranking. The two-stage retrieve-then-rerank paradigm (FAISS [6] + classifier) is standard in dense retrieval. Our key contribution is recognising that the *reranker must be trained on the same similarity distribution as the candidate set it will score*, not on the full pair distribution.

3 Methodology

3.1 Multimodal Encoder Fine-tuning

We fine-tune two separate 768-dimensional backbone encoders with a symmetric InfoNCE objective on same-label pairs from the GLAMI training set:

Text tower: `xlm-roberta-base`, CLS token. We concatenate title and description into a single sequence (max length 128 tokens), which exposes the model to the full multilingual signal available per listing rather than just the title.

Image tower: ImageNet-pretrained `convnext_tiny`, with the classifier head replaced by an identity. Inputs are augmented (random resized crop, horizontal flip, colour jitter) at train time only.

Both towers carry a two-layer projection head ($768 \rightarrow 768 \rightarrow 256$ with GELU) used *only during training*: at inference we extract L2-normalised 768-dimensional backbone features directly, bypassing the projection. This decouples the dimensionality of the contrastive objective from the dimensionality of the embedding consumed by FAISS and the pair classifier.

Training. Each minibatch is constructed from $B = 48$ same-label anchor-positive pairs; all $2(B - 1)$ remaining items act as in-batch negatives. We use gradient accumulation over 4 steps (effective batch 192, giving 382 in-batch negatives per anchor), AdamW with weight decay 0.01, mixed precision, a 10%

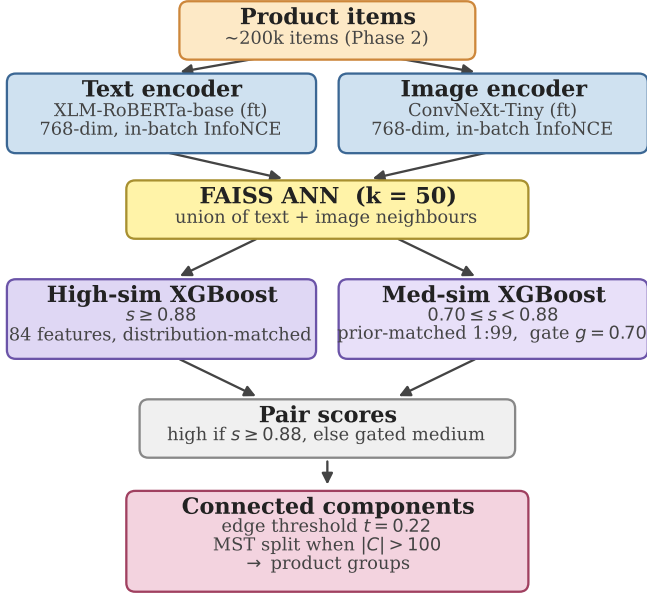


Figure 1: System overview. Fine-tuned text and image encoders feed a FAISS union-of-neighbours retrieval stage. Candidate pairs are split by combined similarity s : a high-similarity classifier (trained on $s \geq 0.88$) handles the dense region, while a separate medium-similarity classifier trained at the inference positive prior ($\sim 1\%$) handles $s \in [0.70, 0.88)$ behind a strict confidence gate. Per-pair scores are merged and connected components produce the final product groups.

linear warmup, and a $10\times$ larger learning rate on the projection head than on the backbone (text: 2×10^{-5} backbone, 2×10^{-4} projection; image: 5×10^{-5} , 5×10^{-4}). InfoNCE temperature $\tau = 0.05$. We train the text tower for 3 epochs and the image tower for 5; the towers are trained sequentially to fit within 12 GB of VRAM (RTX 4070).

We deliberately use only in-batch negatives at the encoder stage, leaving precision to the distribution-matched pair classifier (§3.3–3.4) where hard-negative mining via FAISS does occur. An earlier version replaced both encoders with larger backbones (multilingual-e5-large + dinov2-large, 1024-dim) trained with two-phase ANCE-style hard-negative refinement; this configuration underperformed the simpler in-batch setup on local validation, suggesting that hard negatives in the encoder loss collapse the embedding geometry around borderline cross-geo pairs.

3.2 Candidate Generation

For each item we retrieve the top-50 text neighbours and top-50 image neighbours separately using FAISS IVF (NLIST=500, NPROBE=25, inner product metric). The two neighbour sets are unioned to form candidate pairs. Using both modalities is critical: cross-geo pairs with different language titles have low text similarity but moderate image similarity, so a text-only index would miss them as candidates.

The combined similarity $s = 0.5 \cdot s_{\text{text}} + 0.5 \cdot s_{\text{img}}$ is pre-computed for each candidate pair. All pairs with $s < 0.88$ are discarded; the typical Phase 2 item has ~ 20 – 60 surviving candidates.

3.3 Distribution-Matched Pair Classifier

Candidate pairs after FAISS filtering have $s \geq 0.88$ by construction. A broadly-trained XGBoost classifier assigns near-1.0 probability to every such pair because it has seen the full training distribution (s ranging from 0 to 1), where $s \geq 0.88$ correlates almost perfectly with same-label in the training data. Stated more precisely: trained on the marginal $P(y | \mathbf{x})$, the model has little incentive to learn anything beyond “is s high?” — the bulk of the loss can be driven down using s alone, since the high- s region is overwhelmingly positive in the training pool. At inference, however, every candidate already satisfies $s \geq 0.88$, so this dominant feature is non-discriminative and the model degenerates.

Key insight: we retrain the XGBoost classifier on *exactly the* $s \geq 0.88$ region, where both positives and negatives are abundant (hard negatives share a brand, department, or similar price but belong to different products). This forces the model to learn the conditional $P(y | \mathbf{x}, s \geq 0.88)$, which is the only distribution that exists at inference, and to rely on fine-grained features (PCA differences, exact-match indicators, price ratio) rather than s alone.

Features (84 total): 20 base features including text cosine similarity, image cosine similarity, L2 distances, element-wise difference statistics (mean/std/max for both modalities), combined similarity s , $|s_{\text{text}} - s_{\text{img}}|$, price ratio, price difference, and exact-match indicators for department, colour, and brand; plus 32 text PCA difference features and 32 image PCA difference features (PCA fit on 50k training items, dim=32 each).

Training: 80k positives and 240k negatives mined from the training set (all with $s \geq 0.88$), 1500-estimator XGBoost with early stopping. Val AUC = 0.9990, Val F1 = 0.9884 at $t = 0.30$.

3.4 Recovering Medium-Similarity Pairs

The high-similarity classifier covers candidate pairs with $s \geq 0.88$, but recall analysis on local validation (Section 5) shows that this region contains only $\sim 90.6\%$ of the true-positive pairs surviving FAISS. The remaining $\sim 8\%$ sit in $s \in [0.70, 0.88]$ – almost exclusively cross-geo pairs whose titles share no subword tokens and whose images differ in crop and background. They are reachable by FAISS but unreachable by a classifier never trained to discriminate them.

We train a second XGBoost on this medium-similarity range with the same 84 features. The principle from §3.3 applies, but with an additional twist: the medium range has a very different positive prior. In $s \in [0.70, 0.88]$ the empirical positive rate is $\sim 1\%$, two orders of magnitude below the ≥ 0.88 region. A naive training mixture (e.g., 1:3 negatives) is therefore mismatched not just in similarity but in *class prior*, so the model systematically over-predicts at inference. We mine 60k positives and 5.94M negatives (1% positive rate, matching inference), train without metadata-based hard-negative filters since pairs in this range are inherently hard, and apply the model with a strict confidence gate: a medium-similarity pair is promoted only when its score exceeds $g = 0.70$. Below the gate, the pair receives score 0. The gate is necessary because the model’s predictions are well-calibrated only at high confidence in this prior regime; relaxing it injects false positives whose cluster-level damage (through bridging edges) outweighs the recall gain (Section 5).

3.5 Clustering

We threshold pair scores at $t = 0.22$ and build a sparse adjacency graph. Connected components yield product groups. Any component exceeding 100 items (the submission constraint) is split by binary search on an MST weight cut.

4 Baselines

Lexical baselines (exact title, TF-IDF). We do not report numbers for exact-title matching or TF-IDF cosine similarity because they are structurally incapable of solving the task: cross-geo pairs share virtually no surface tokens (e.g. a Czech and a Romanian title for the same dress have zero string overlap and almost zero subword overlap). Any lexical baseline collapses to roughly the same-language subset of pairs, leaving cross-language matches — the dominant difficulty of this dataset — entirely unrecovered. This motivates the multilingual encoder fine-tuning in Section 3.1.

Cosine threshold (raw similarity). Using the fine-tuned combined similarity s directly, the best threshold on Phase 2 local validation is $t = 0.95$, yielding Pairwise F1 = 0.8796. This is the strongest naive baseline and the one we improve upon.

Broadly-trained XGBoost. Training the same 84-feature XGBoost on the full pair distribution (not restricted to $s \geq 0.88$) produces the calibration failure described in Section 3.3: the model assigns score ≈ 1.0 to virtually all FAISS candidates, collapsing

the connected components into one giant cluster ($F1 < 0.10$). This failure mode demonstrates that distribution matching is not an optimisation trick but a correctness requirement.

5 Experiments

Phase 1. On the 15k predefined groups of 5 items from `items_phase1.csv`, our pair classifier scores all $\binom{5}{2} = 10$ pairs per group. A group is predicted positive if any pair exceeds $t = 0.22$, the same threshold used for Phase 2 clustering. Phase 1 F1 = **0.9937**, confirming that the classifier reliably detects duplicates when they are present and that no Phase-1 specific tuning is required.

Local validation for Phase 2. We construct a Phase-2 proxy from labelled training items that are disjoint by product label from those used to fit the XGBoost models and embedding PCAs (185k items, 30k labels, 1.09M true-positive pairs), then run the full FAISS $\rightarrow$ classifier $\rightarrow$ connected-components pipeline against their ground-truth labels. Splitting by label rather than by item ensures no positive pair leaks between training and validation.

Recall budget. Before reporting F1 numbers, we measure recall at each pipeline stage on the validation set. FAISS retrieval (top-50 union of text and image neighbours) recovers 98.5% of true-positive pairs. Restricting to $s \geq 0.88$ – the high-similarity classifier’s training region – drops recall to 90.6%; the missing 7.9 percentage points sit in $s \in [0.70, 0.88]$, which is exactly what the medium-similarity classifier is designed to recover.

Threshold sweep (Phase 2). Figure 2 and Table 1 report local-validation F1 at selected thresholds for the full system (high-similarity classifier plus distribution-matched medium-similarity classifier with gate $g = 0.70$).

Table 1: Local-validation F1 at selected thresholds. *Base*: high-similarity classifier only ($s \geq 0.88$), best at $t = 0.28$. *+medmatch*: full system including distribution-matched medium-similarity classifier with gate $g = 0.70$, best at $t = 0.22$. “–” indicates the value was not in the original Base sweep.

Threshold t	Base F1	+medmatch F1
0.20	0.9004	0.9258
0.22	–	0.9260
0.26	0.9042	0.9253
0.28	0.9044	0.9252
0.30	0.9043	0.9243
0.34	0.9038	0.9233
0.40	0.9019	0.9209
0.50	0.8966	0.9173
0.95	0.8796	0.8799

The full-system F1 curve plateaus between $t = 0.20$ and $t = 0.32$ ($F1 \in [0.9243, 0.9260]$), with the maximum at $t = 0.22$. Adding the medium-similarity classifier improves F1 by approximately 2.2 points across the entire plateau, with the largest gains at lower thresholds where the recovered medium-similarity pairs survive clustering.

Gate sensitivity. We swept the medmatch gate $g \in \{0.50, 0.70, 0.80\}$ on local validation. Lowering g to 0.50 pro-

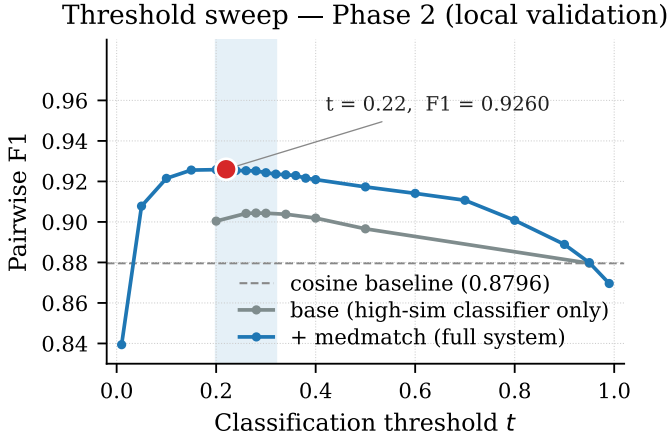


Figure 2: Pairwise F1 vs. classification threshold on local validation. The dashed line marks the cosine-similarity baseline (F1 = 0.8796). The shaded band shows the stable plateau; we operate at $t = 0.22$.

moted 20.9k pairs (vs. 10.1k at $g = 0.70$) and recovered 11.4k additional true positives, but added 25.6k false positives, dropping best F1 to 0.9207. The 1%-prior medmatch model is well-calibrated only at high confidence; below $g = 0.70$, false-positive bridging edges connect distinct clusters and the cluster-level damage outweighs the recall gain. We therefore operate at $g = 0.70$.

Failed extensions. Two further extensions failed to improve over the system above:

Naive medium-similarity model. An earlier medium-similarity XGBoost trained on a 25% positive mixture (1:3 negatives) with brand- or department-matched hard negatives gave local F1 = 0.27 – a second instance of the distribution-mismatch failure mode (§3.3, §4), here in the class prior. The fix in §3.4 (prior-matched sampling and strict gating) turns the same architecture into a +2.2 F1 improvement.

Anchor bridging. Linking phase items via shared high-similarity training-set anchors generated false bridges (e.g. two distinct red dresses resembling the same training dress), reducing F1 below the cosine baseline.

Rule-based metadata promotion. Pairs with brand + department + colour exact match in $s \in [0.70, 0.88]$ were promoted, but brand and department IDs are near-zero for most items, so only 5 pairs qualified across the full inference set.

6 Results

Our best configuration achieves Pairwise F1 = **0.9260** on local validation at $t = 0.22$, compared to the cosine-threshold baseline of 0.8796 (+0.046). The high-similarity classifier alone (without medmatch) achieves 0.9044 at $t = 0.28$; adding the distribution-matched medium-similarity classifier with strict gating contributes the remaining +0.022. Phase 1 F1 = 0.9937 with no Phase-1 specific tuning.

Error analysis. Two failure modes dominate the remaining false negatives. First, $\sim 1.5\%$ of true positives never enter the candidate set: FAISS retrieval recall caps at 98.5% even with top-50 union of text and image neighbours. Second, the 1.0% of

true positives that lie below $s = 0.70$ (and are in the candidate set) are invisible to both classifiers, since neither operates below that similarity. Both groups are dominated by cross-geo pairs where titles share no subword tokens and product photos have different backgrounds or cropping conventions. The fine-tuned encoders reduce intra-language variance but do not fully align cross-language representations.

Ablation. Two ablations isolate the contribution of distribution matching. *First*, replacing the high-similarity classifier with a broadly-trained XGBoost (identical features and hyperparameters, but trained on the full $s \in [0, 1]$ range) collapses the system entirely (F1 < 0.10), because the model assigns near-1.0 probability to every FAISS candidate and connected components merge into one giant cluster. *Second*, replacing the prior-matched medium-similarity classifier with the naive 25%-positive variant (Section 5) gives F1 = 0.27 – the same architecture and features fail catastrophically when the training positive prior does not match inference. Both ablations point to the same underlying mechanism: when training and inference distributions disagree (in similarity range, in class prior, or both), no amount of feature engineering or model capacity recovers the calibration loss.

7 Conclusion

We presented a two-stage system for cross-geo product grouping: fine-tuned multimodal encoders followed by two distribution-matched XGBoost pair classifiers. The central finding is that rerankers trained on the full pair distribution are miscalibrated when applied to FAISS candidates, and that the same principle extends to the class prior: a medium-similarity classifier trained at 25% positives fails on a 1%-positive inference set, but training at the matched prior with strict confidence gating recovers +2.2 F1 points. The full system reaches 0.9260 Pairwise F1 on local validation, +0.046 over the cosine baseline.

The remaining gap stems from (i) the 1.5% of true-positive pairs that FAISS fails to retrieve even with text+image union, and (ii) the 1.0% of true positives below $s = 0.70$, dominated by cross-geo pairs with no shared subword tokens. Promising avenues for future work are stronger multilingual alignment in the text encoder (e.g. language-pair supervision) and adding a cross-encoder score as a feature into the medium-similarity classifier, since cross-encoders capture textual semantics that bi-encoder cosine similarity cannot.

References

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